

THUMMALAPALLI HARSHA

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PROFESSIONAL SUMMARY

Computer Science graduate with hands-on experience in Machine Learning, Data Science, and AI/GenAI through academic projects and internship experience. Skilled in Python, Scikit-learn, SQL, exploratory data analysis, classification, and model evaluation, with additional experience building RAG and LLM-powered applications. Developed practical projects in fraud detection, student-data retrieval, and natural-language SQL generation. Seeking an entry-level opportunity to apply machine learning, data analysis, and AI skills to real-world problems.

SKILLS

Programming: Python, SQL

Libraries / Frameworks: Pandas, NumPy, Scikit-learn, Streamlit

AI / Machine Learning: RAG, Embeddings, Generative AI, Ollama, XGBoost, SMOTE, Classification, Regression

Databases: SQLite, SQL, ChromaDB

APIs / Integration: Google Gemini API, API integration, Requests

Tools / Platforms: Git

Languages: English, Telugu, Hindi, Tamil

PROJECTS

SmartFraud Classifier

- Loaded and explored a 284,807-transaction fraud dataset, applying preprocessing and class-imbalance handling with SMOTE.
- Compared Decision Tree, Random Forest, and XGBoost models for binary fraud classification.
- Evaluated model performance with precision, recall, and F1 metrics and saved model artifacts.

Tech Stack: Python, Pandas, NumPy, Scikit-learn, XGBoost, Matplotlib, Seaborn, Imbalanced-learn

Project Link: <https://github.com/thummalapalliharsha/SmartFraud-Classifer>

Student Performance RAG Chatbot

- Built a local RAG chatbot over 6,607 student records using preprocessing and searchable student profiles.
- Generated embeddings, stored 6,607 student documents in ChromaDB, and performed semantic retrieval.
- Integrated Ollama with llama3.2:3b and used Langfuse tracing, Promptfoo evaluation, and Pytest testing.

Tech Stack: Python, Pandas, Streamlit, Ollama, llama3.2:3b, nomic-embed-text, ChromaDB, Langfuse

Project Link: <https://github.com/thummalapalliharsha/student-performance-rag>

Text-to-SQL Project

- Built a Python workflow that converts natural-language questions into SQLite-compatible SQL queries.
- Integrated a Gemini REST API with an OpenAI-compatible fallback and executed generated SQL against a customers/orders database.
- Implemented structured API payloads, code-fence removal, first-statement handling, and query-result return.

Tech Stack: Python, SQLite, SQL, Google Gemini API, OpenAI-compatible client, Requests, Python-dotenv

Project Link: https://github.com/thummalapalliharsha/text_to_sql_project

EXPERIENCE

Artificial Intelligence Intern — Ediglobe (February 2024–March 2024)

- Implemented data preprocessing pipelines in Python.
- Built and evaluated classification models using Scikit-learn.
- Performed exploratory data analysis on structured datasets.

EDUCATION

Bharath Institute of Higher Education and Research — B.Tech

Computer Science and Engineering (2022–2026); CGPA: 7.94

Harvest Public School — Class XII — CBSE (2022); 70.8%

Harvest Public School — Class X — CBSE (2020); 82.2%

CERTIFICATIONS

- AI-ML Virtual Internship — EduSkills (March 2025)
- Artificial Intelligence Internship — Ediglobe (February 2024)
- Altair RapidMiner Data Science Master — AICTE (June 2025)
- TCS iON Career Edge – Young Professional — TCS iON (March 2025)